GLUCOTWIN

SYSTEM ARCHITECTURE | DIGITAL TWIN FOR TYPE 2 DIABETES

Dual-stream synthetic data → feature fusion → risk estimates → clinician review

STREAM A — SYNTHETIC EHR Demographics • diagnoses • HbA1c/eGFR • medications • comorbidities	**STREAM B — SIMULATED WEARABLES** CGM • HRV • sleep • steps • recent trends
VALIDATION & FEATURE ENGINEERING **Quality checks • time windows • missingness handling**	**PERSISTENT TWIN STATE** **Recent trends • risk history • last-test dates • clinician actions**
EARLY FEATURE FUSION **Combine sensor features with EHR context**	**VERSIONED RULE ENGINE** **Investigation review prompts • provenance • clinician review**
LIGHTGBM MODEL PIPELINE **Hyperglycaemia & hypoglycaemia estimates** **Nominal 120-minute horizon; experimental**	**EXPLANATION LAYER** **Feature-contribution summaries where configured**
FASTAPI SERVICE **Typed endpoints • prediction/context payloads • audit events**	**CLINICIAN REVIEW & AUDIT** **Dashboard • accept/defer/override logging**

Flow: both streams are validated and engineered, fused for risk estimation, then exposed through the API to a clinician-facing dashboard. Twin state and audit events support review over time.

Safety boundary: outputs are review prompts only. The prototype does not diagnose, prescribe, order/cancel investigations, or change therapy. All demo data is synthetic; clinical rules and thresholds require expert review.

Components and responsibilities

Layer	Responsibility	Technology / artifact
Data	Create synthetic clinical context and sensor-like time series.	Python generator; NumPy; SciPy; Pandas; Parquet
Features	Validate, window, derive and fuse sensor + EHR features.	Pandas; project feature pipeline
Model	Estimate event probabilities over a nominal 2-hour horizon.	LightGBM; scikit-learn; evaluator
Explainability	Expose contribution explanations where configured.	SHAP / model explanation layer
Twin state + rules	Maintain trends and present versioned investigation prompts.	SQLite state; rule files; audit log
Application	Serve prediction/context and present clinician review workflow.	FastAPI; Pydantic; React; TypeScript; Vite

Evaluation integrity note

Current generated headline metrics and ablation CSV disagree (hypoglycaemia AUROC: 0.5000 in metrics.json versus 0.8264 for full fusion in ablation.csv). Reconcile the evaluation implementation before making performance claims. Synthetic-data metrics do not establish clinical performance.